Huawei Access Node IND-ACC-58 Failure Response Procedure (ALARM_9206)

Node: IND-ACC-58 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_9206

Description

The failure domain of Hardware/Environmental Fault on Huawei Access Nodes is a complex phenomenon that can arise due to a multitude of factors. In the context of Node ID IND-ACC-58 at the Indiranagar Exchange, this failure can be attributed to ASIC memory parity errors or line card crashes, which are critical hardware components responsible for processing and forwarding traffic. The loss of power supply redundancy further exacerbates the situation, rendering the node vulnerable to complete system failure. This is particularly concerning as the Access Node serves as a critical juncture in the network, providing connectivity to a vast array of devices and services. The Huawei Access Node's robust architecture, incorporating advanced features such as ASIC-based processing and redundant power supply units, is designed to mitigate such failures. However, when these components fail, the node's ability to function is severely impaired, resulting in the ALARM_9206 being triggered. In this SOP, we will outline the procedures for diagnosing and resolving this critical issue, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- step: Verify Power Supply Redundancy, description: Ensure that the power supply units are functioning correctly and that redundancy is maintained.

- step: Inspect Line Card Status, description: Verify that the line cards are functioning correctly and that there are no signs of physical damage or malfunction.

- step: Check Environmental Conditions, description: Verify that the environmental conditions at the Indiranagar Exchange are within the recommended operating parameters for the Huawei Access Node.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display fault-management error-alarm, Expected Output:
ALARM_9206: ASIC memory parity error detected on Line Card 1,
Description: This command retrieves the current error-alarm
information, providing insight into the root cause of the failure.
```

```
Command: display device power-supply, Expected Output: Power Supply 1: Normal, Power Supply 2: Faulty, Description: This command displays the current power supply status, enabling identification of any issues with redundancy.

Command: display line-card status, Expected Output: Line Card 1: Faulty, Line Card 2: Normal, Description: This command retrieves the current status of the line cards, allowing for identification of any faulty components.
```

Resolution Steps

Step 1: Step: Replace Faulty Line Card, Command: replace line-card 1, Description: This step involves replacing the faulty line card with a spare unit, ensuring that the node's functionality is restored.

Step 2: Step: Repair or Replace Power Supply, Command: repair power-supply 1, Description: This step involves repairing or replacing the faulty power supply unit, ensuring that redundancy is maintained and the node's functionality is restored.

Step 3: Step: Clear ASIC Memory Parity Errors, Command: clear asic-memory-parity-error, Description: This step involves clearing the ASIC memory parity errors, restoring the node's functionality.

Post-Check Verification

```
Command: display device status, Expected Output: Device Status: Normal, Description: This command verifies that the node's functionality has been restored and that all components are functioning correctly.

Command: display line-card status, Expected Output: Line Card 1: Normal, Line Card 2: Normal, Description: This command verifies that the line cards are functioning correctly and that redundancy is maintained.
```

Roll-back Procedures

- Step: Revert Line Card Replacement, Command: undo replace line-card 1, Description: This step involves reverting the replacement of the line card, restoring the faulty unit to the node.

- Step: Revert Power Supply Repair or Replacement, Command: undo repair power-supply 1, Description: This step involves reverting the repair or replacement of the power supply unit, restoring the faulty unit to the node.

- Step: Clear ASIC Memory Parity Errors, Command: undo clear asic-memory-parity-error, Description: This step involves clearing the ASIC memory parity errors, restoring the node's functionality to its original state.